

Exploring Place Value and Number Size

SimuPhysics Place Value Slider · SimuPhysics

Years 7-10

~30 minutes

Mathematics · Number

NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://simuphysics.com/view/mathematics-place-value-slider.html>

Curriculum links: AC9M7N03 · AC9M10N01

Learning intentions

- I can build a number on a place-value chart and explain what each digit represents.
- I can write numbers in expanded notation using powers of 10.
- I can explain what happens to a number's value when a digit moves between columns.

Getting started

Open the simulation. Start by setting a small target number (e.g. a 3-digit number) and slide the digits into the correct columns on the place-value chart to build it.

Tasks

1. Build your target number. Which column did you place the first digit in, and what value does that digit represent in that column?

2. Slide the same digit one column to the left. What is its new value, and by what factor did its value change?

3. Write your target number in expanded notation, showing each digit multiplied by the correct power of 10 (e.g. $3 \times 100 + 4 \times 10 + 5 \times 1$).

Working:

4. Build a larger target number, at least 5 digits, and write it in expanded notation the same way as task 3.

Working:

5. Move the same digit through three different columns on the chart and record its resulting value each time.

Column digit is placed in	Value of the digit in that column

6. Using your table, explain in your own words why the same digit can represent very different values depending on where it sits in a number.

7. A digit is accidentally placed one column too far to the right. Explain what effect this has on the size of the resulting number.

Extension (optional)
Large numbers are often rounded for convenience (e.g. rounding 4,982,000 to 5,000,000). Use the place-value chart to help explain why this kind of rounding can make a bigger difference to accuracy when the number is very large.